

PySpark Wine Reviews: End to End Big Data ML

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Team Name: Society of Wine Specialists

1. Introduction and Motivation

The luxury consumer goods market represents a complex domain where understanding consumer preferences and predicting market value requires sophisticated analytical approaches. Wine, as a high-end consumer product, presents unique challenges due to the multidimensional nature of quality assessment, which combines objective characteristics (price, origin, vintage) with subjective expert evaluations (tasting notes, point scores).

This project addresses the fundamental question:

How accurately can we predict high-end consumer goods pricing and characteristics using machine learning techniques?

Using a comprehensive wine review dataset with approximately 120,000 entries, we aim to understand the complex relationships between expert ratings, descriptive language, geographical origin, and market pricing. The objective is to develop fine-tuned predictive models that can accurately forecast consumer taste preferences and market values in the luxury goods sector.

This problem inherently benefits from big data approaches for several critical reasons:

- **High-dimensional text analysis:** Processing thousands of wine descriptions requires TF-IDF vectorization resulting in ~8,000 dimensions, necessitating distributed computing for dimensionality reduction via PCA
- **Large-scale feature engineering:** Combining numerical, categorical, and text features across 120,000 records demands scalable preprocessing pipelines
- **Computational complexity:** Training ensemble models like Random Forest on high-dimensional data requires parallel processing to achieve reasonable execution times
- **Iterative model experimentation:** Testing multiple PCA configurations and model architectures in a cluster environment enables systematic performance optimization

2. Project Objectives

The project encompasses the following specific objectives:

- **Comprehensive exploratory data analysis (EDA):** Analyze country distribution, grape variety characteristics, vintage trends, and the relationship between price and expert ratings across approximately 120,000 wine reviews
- **Scalable data preprocessing pipeline:** Develop a 14-step standardized preprocessing workflow using PySpark that handles missing values (61%

region_2 missing), removes duplicates (9,985 records), performs outlier detection, and creates reusable transformation pipelines

- **Advanced feature engineering:** Implement TF-IDF text vectorization on wine descriptions, apply PCA dimensionality reduction (from ~8k to ≤500 dimensions), and encode categorical variables using StringIndexer and OneHotEncoder
- **Multi-task machine learning implementation:** Develop and compare regression models for price prediction (Linear Regression, Random Forest), multi-class classification for variety prediction, and text-based classification for origin prediction
- **Cluster performance evaluation:** Conduct 6 systematic experiments on ICDS/Roar-Collab infrastructure testing PCA's impact on computational performance, memory pressure, and compression efficiency across different node configurations (2-node vs 4-node Spark clusters)
- **Demonstrate data leakage prevention:** Contrast model performance with and without label leakage to emphasize best practices in feature selection and model validation

3. Data Description

Data Source

Kaggle WineEnthusiast dataset, a public repository containing comprehensive wine reviews from professional critics.

Data Size

Approximately 130,000 records initially; after cleaning and deduplication, 119,927 valid records. The dataset includes 13 fields covering wine characteristics, expert evaluations, and metadata.

Data Characteristics

- **Structured data:** Numerical fields (price, points), categorical fields (country, province, variety, winery)
- **Unstructured text:** Long-form wine descriptions requiring natural language processing
- **Temporal dimension:** Vintage years extracted from wine titles enabling time-series trend analysis
- **Geographical hierarchy:** Multi-level location data (country → province → region_1 → region_2)

Data Challenges

- **Significant missing values:** 61% of region_2 entries missing, requiring strategic imputation or exclusion decisions
- **Substantial duplicates:** 9,985 duplicate records identified through content fingerprinting (title + winery + description hashing)
- **Skewed price distribution:** Extreme outliers requiring 1%-99% percentile clipping; non-linear relationship with ratings necessitating log transformation
- **High dimensionality:** TF-IDF vectorization generates approximately 8,000 features, creating computational challenges and requiring PCA dimensionality reduction

- **Categorical cardinality:** Hundreds of unique countries, provinces, and varieties requiring efficient encoding strategies
- **Label leakage risk:** Geographical fields in features could directly reveal target variables in classification tasks, requiring careful feature selection

4. Technical Approach

Tools & Frameworks

- **PySpark:** Primary framework for distributed data processing, utilizing DataFrames for structured operations and MLlib for machine learning pipelines
- **MLlib components:** StringIndexer, OneHotEncoder for categorical encoding; HashingTF, IDF for text vectorization; PCA for dimensionality reduction; LinearRegression, RandomForestRegressor, LogisticRegression for modeling
- **Visualization:** Matplotlib for generating charts and graphs during exploratory analysis
- **Data formats:** Parquet for efficient columnar storage; CSV for initial data ingestion

Cluster Usage Plan

- **Infrastructure:** ICDS Roar-Collab distributed computing platform
- **Node configurations:** Experiments conducted on 2-node and 4-node Spark clusters to evaluate scalability
- **Job types:** Batch processing for data cleaning and feature engineering; iterative ML training for model development; performance benchmarking across different PCA dimensions (50/100/200/500)
- **Storage requirements:** Two output datasets—wine_clean.parquet for EDA, wine_ml_ready.parquet for modeling; persistent storage of trained preprocessing pipelines and PCA models for reusability
- **Performance testing:** 6 systematic experiments measuring execution time, memory consumption, and compression efficiency across different partitioning strategies and PCA dimensions

Pipeline Overview

Phase 1: Data Ingestion and Preprocessing

- CSV loading with special handling for multi-line text and quote escaping
- Data type correction: converting points to integers, cleaning price commas and converting to floats
- Missing value analysis and strategic imputation using country+variety medians
- Text standardization: lowercase conversion, accent removal, whitespace normalization
- SHA-256 content fingerprinting for duplicate detection
- Outlier treatment using 1%-99% percentile boundaries
- Grape variety synonym consolidation (e.g., Pinot Gris/Pinot Grigio normalization)

Phase 2: Feature Engineering

- TF-IDF text features: Tokenization, stopwords removal, term frequency-inverse document frequency calculation generating ~8,000-dimensional vectors

- PCA dimensionality reduction: Experimentation with 50, 100, 200, and 500 principal components to balance information retention with computational efficiency
- Categorical encoding: StringIndexer for label encoding followed by OneHotEncoder for country, variety, and province variables
- Feature assembly: VectorAssembler combining numerical features (points, price), encoded categorical features, and PCA-reduced text embeddings into unified feature vectors

Phase 3: Model Building and Evaluation

Task 1: Price Prediction (Regression)

- Baseline Linear Model: Using only text embeddings and point scores
- Enhanced Linear Model: Adding geographical and variety features
- Random Forest Model: Capturing non-linear feature interactions
- Evaluation metrics: RMSE, R^2 score, practical testing on new wine descriptions

Task 2: Variety Classification (Multi-class)

- Logistic Regression using text features, ratings, and geographical context
- Predicting grape varieties from tasting notes and origin information

Task 3: Origin Prediction with Leakage Analysis

- With-leakage model: Including direct geographical features (accuracy $\approx 99.8\%$)
- Without-leakage model: Using only tasting notes, scores, and vintage (accuracy $\approx 74.6\%$)
- Demonstrates the importance of rigorous feature selection to avoid inflated performance metrics

Phase 4: Visualization and Interpretation

- Country distribution analysis with pie charts and bar graphs
- Variety-wise price and rating distributions
- Vintage trend analysis showing temporal patterns
- Price-rating correlation studies including log-transformation revealing logarithmic relationships
- Binned statistical analysis exposing non-linear patterns in premium wine pricing

5. Results and Discussion

Data Visualization Insights

Country and Variety Distribution

The exploratory analysis revealed that the United States dominates the dataset with 50,456 wines (approximately 42% of all records), followed by France and Italy. Regarding grape varieties, Pinot Noir, Chardonnay, and Cabernet Sauvignon emerge as the most frequently reviewed, reflecting their commercial prominence in global wine markets.

Price-Rating Relationship Discovery

A critical finding from the EDA phase is the logarithmic relationship between price and expert ratings. The original price-rating correlation was only 0.40, but after log-transformation, the correlation coefficient increased to 0.59, indicating a much stronger linear relationship in log space. Linear regression in log space yielded the formula: $\log(\text{price}) \approx 0.1225 \times \text{points} - 7.52$, suggesting that each 1-point increase in

rating corresponds to approximately 13% price increase on average.

Furthermore, binned analysis revealed distinct pricing tiers:

- Wines scoring 80-86 points cluster in the \$16-22 range with low variance
- 87-90 point wines show price escalation to \$30-40 average
- 92+ point "trophy wines" exhibit extreme price dispersion, with some reaching several hundred dollars while maintaining similar ratings

This pattern underscores that expert ratings are significant but incomplete predictors of price—brand prestige, rarity, and terroir reputation contribute substantially to luxury wine pricing.

Machine Learning Model Performance

Price Prediction Results

Three regression models were systematically compared:

- **Baseline Linear Model:** Achieved $R^2 \approx 0.275$ and $RMSE \approx 31.3$ using only TF-IDF text features and point scores. This modest performance demonstrates that tasting notes alone provide meaningful but limited pricing information.
- **Enhanced Linear Model:** Incorporating geographical origin (province) and grape variety improved R^2 to 0.321 ($RMSE \approx 30.3$). The incremental gain highlights the importance of provenance and varietal characteristics in wine valuation.
- **Random Forest Model:** Delivered the best performance with $R^2 \approx 0.429$ and $RMSE \approx 27.8$. The substantial improvement over linear models indicates that wine pricing involves complex non-linear interactions between features—textual descriptions, ratings, regions, and varieties combine in intricate ways that tree-based models can capture.

Practical validation using a new Burgundy Pinot Noir description (not in training data) predicted a price of approximately \$65.40, which aligns with typical premium Burgundy pricing, demonstrating the model's real-world applicability.

Origin Prediction and Data Leakage Analysis

This task provided a powerful demonstration of data leakage risks in machine learning:

- **With-leakage model:** Using province and region fields as features achieved near-perfect accuracy (99.8%, $F1 \approx 0.997$). While impressive, this represents cheating—the model simply memorized geographical hierarchies.
- **Without-leakage model:** Restricted to only tasting notes, point scores, and vintage information, accuracy dropped to 74.6% ($F1 \approx 0.695$). This realistic scenario shows that wine descriptions contain subtle linguistic cues—flavor profiles, terroir terminology, and stylistic language—that allow reasonably accurate origin inference.

For instance, an Austrian Grüner Veltliner description was correctly predicted as Austrian origin with 79.9% confidence, based on characteristic descriptors in the tasting notes. This analysis reinforces critical lessons about feature engineering and highlights the danger of inadvertently including target-correlated variables.

Cluster Performance and Scalability

Six systematic experiments were conducted on the ICDS Roar-Collab platform to

evaluate PCA's impact on computational efficiency:

- **Execution time analysis:** Jobs processing the full data pipeline (ingestion → cleaning → TF-IDF/PCA → ML-ready features) took several minutes to over ten minutes depending on PCA dimensionality and partition count. Lower PCA dimensions (50-100) completed faster than higher dimensions (200-500), confirming that dimensionality reduction accelerates downstream processing.
- **Memory pressure testing:** Higher PCA dimensions increased memory consumption during model training. The 500-dimension configuration approached resource limits on 2-node clusters, while the 4-node setup handled it comfortably, demonstrating the scalability advantage of additional computational resources.
- **Compression efficiency:** PCA-reduced datasets stored as Parquet files showed 30-40% size reduction compared to full TF-IDF vectors, improving I/O performance and storage costs.
- **Multi-node scaling:** Comparing 2-node versus 4-node clusters revealed that doubling compute capacity reduced job execution time by approximately 35-45% for PCA-heavy stages, though not perfectly linear due to coordination overhead.

These experiments validate that PCA is not merely a theoretical tool but provides tangible engineering benefits in real-world big data workflows—reducing computation time, alleviating memory constraints, and decreasing storage requirements.

Code and Data Accessibility

All code, trained models, and processed datasets are version-controlled and available in the project's GitHub repository. The repository includes:

- Jupyter notebooks documenting EDA, preprocessing, and ML tasks
- PySpark scripts for reproducible pipeline execution
- Saved preprocessing models (TF-IDF pipelines, PCA models, encoders) for applying to new data
- Cleaned datasets (wine_clean.parquet and wine_ml_ready.parquet)
- Cluster experiment logs and performance benchmarks

GitHub Repository: [Insert your GitHub repository URL here]

Unsolved Problems and Future Directions

Despite achieving meaningful results, several challenges remain unresolved:

- **Limited price prediction accuracy:** Even the best Random Forest model explains only 43% of price variance. Future work could incorporate external features such as critic reputation, vintage ratings from external sources, and brand equity metrics to improve predictive power.
- **Temporal dynamics:** The current analysis treats data as static. Incorporating time-series modeling to capture how wine prices evolve with aging, market trends, and vintage reputation changes over time would add significant value.
- **Advanced NLP techniques:** TF-IDF captures term importance but ignores semantic meaning. Experimenting with word embeddings (Word2Vec, GloVe) or transformer-based language models (BERT) could extract richer information from tasting notes.
- **Cross-domain generalization:** The methodology developed here focuses specifically on wine. Testing whether similar approaches can predict prices

and preferences for other luxury goods (watches, art, spirits) would validate the framework's broader applicability.

- **Interactive recommendation system:** Building a user-facing application that recommends wines based on personal taste profiles (derived from previous ratings and descriptions) would be a practical extension.

If revisiting this project, the primary focus would be on integrating transformer-based NLP models for deeper textual understanding and conducting longitudinal analysis to capture temporal pricing dynamics. Additionally, deploying the models as a web service with a recommendation interface would demonstrate real-world utility beyond academic experimentation.

6. CREDIT Statement

Following the Elsevier CRediT (Contributor Roles Taxonomy) guidelines:

Data acquisition & preprocessing lead: **Kevaghna McClymont and Ce Liang**

PySpark pipeline & ML lead: **Varun Kaveti and Yash Botre**

Visualization & results lead: **Kevaghna McClymont and Abha Mam**

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8. References

Zackthoutt. (2017). *Wine Reviews* [Data set]. Kaggle. Retrieved September 25, 2025, from <https://www.kaggle.com/datasets/zynicide/wine-reviews>